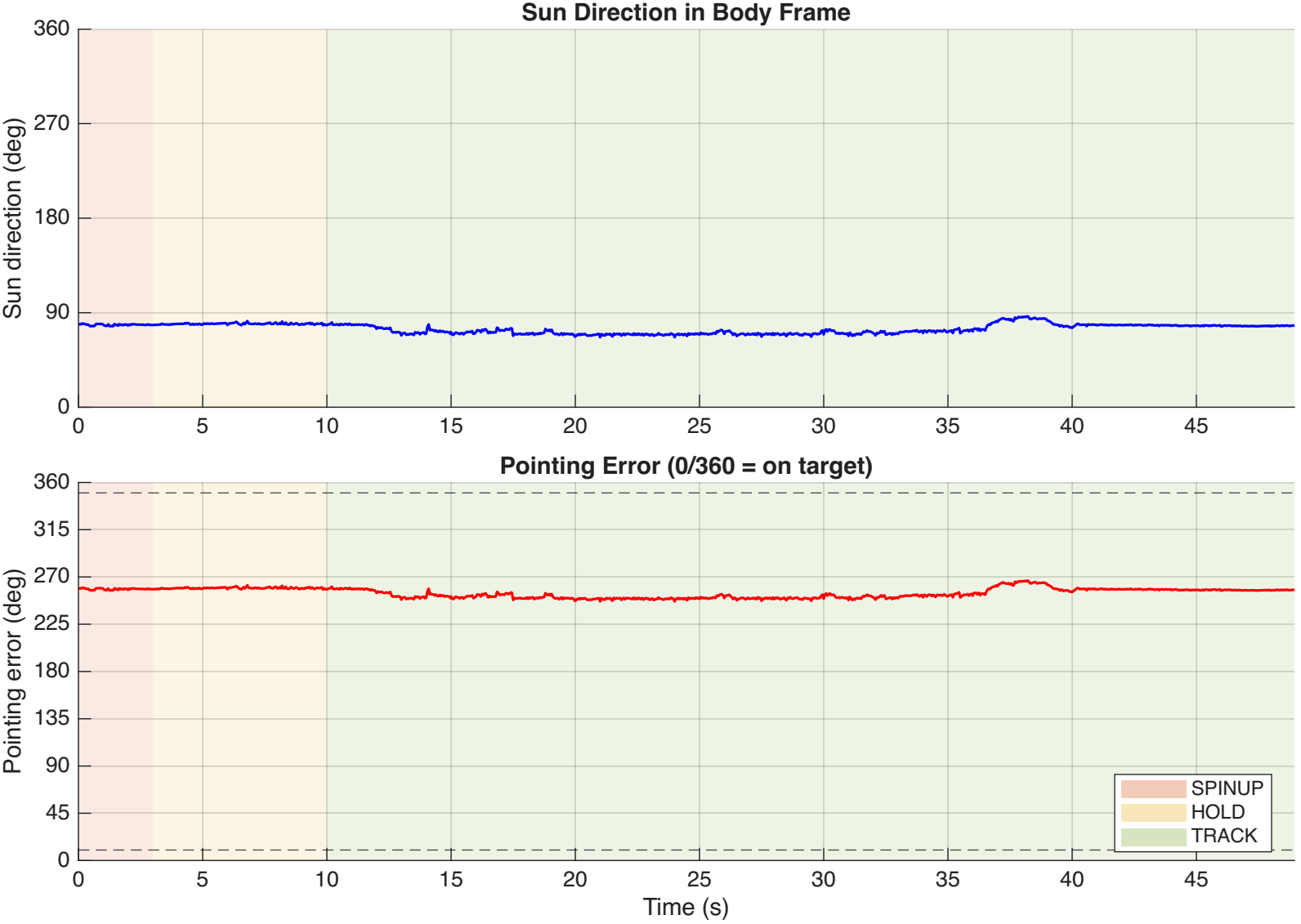
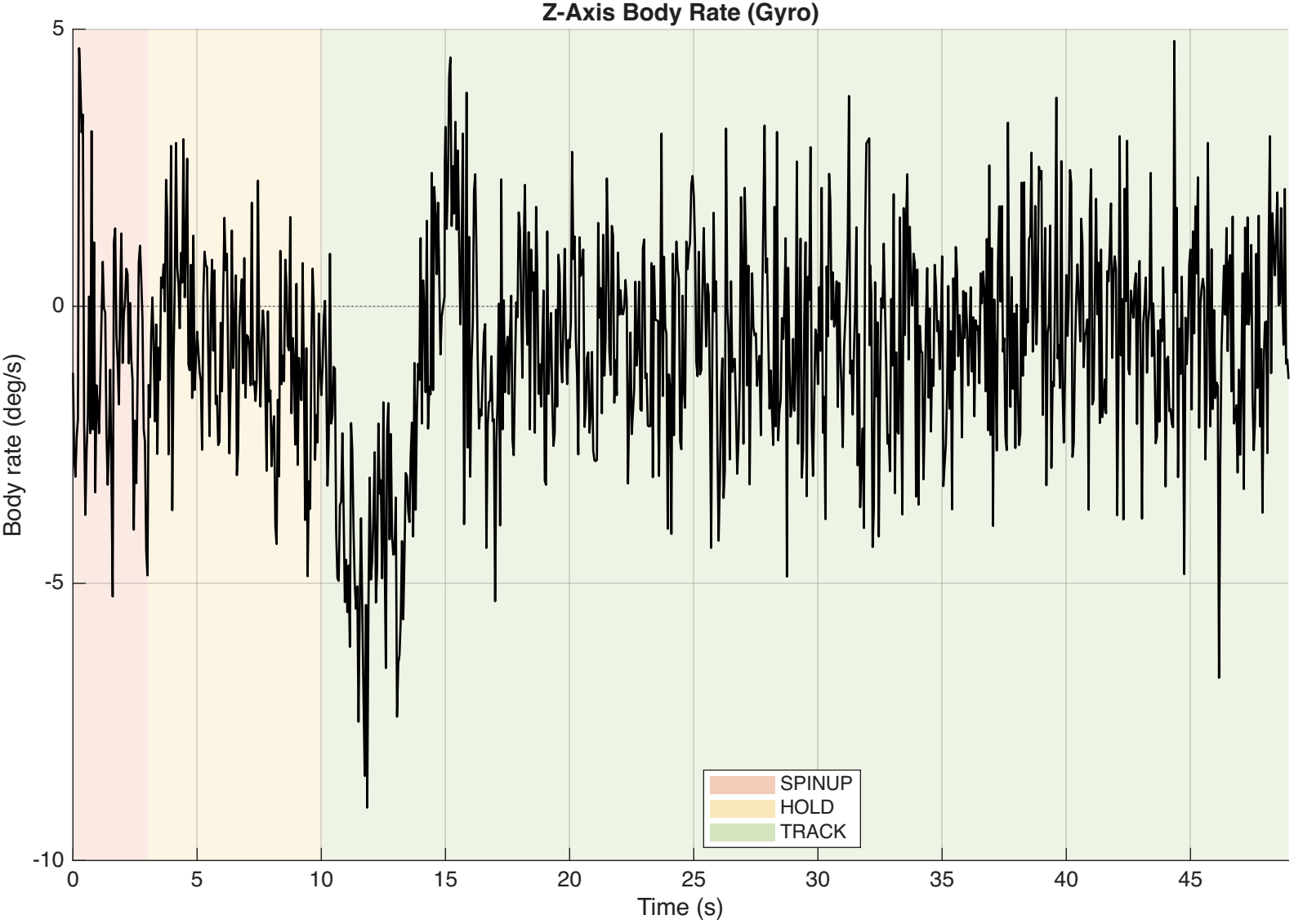
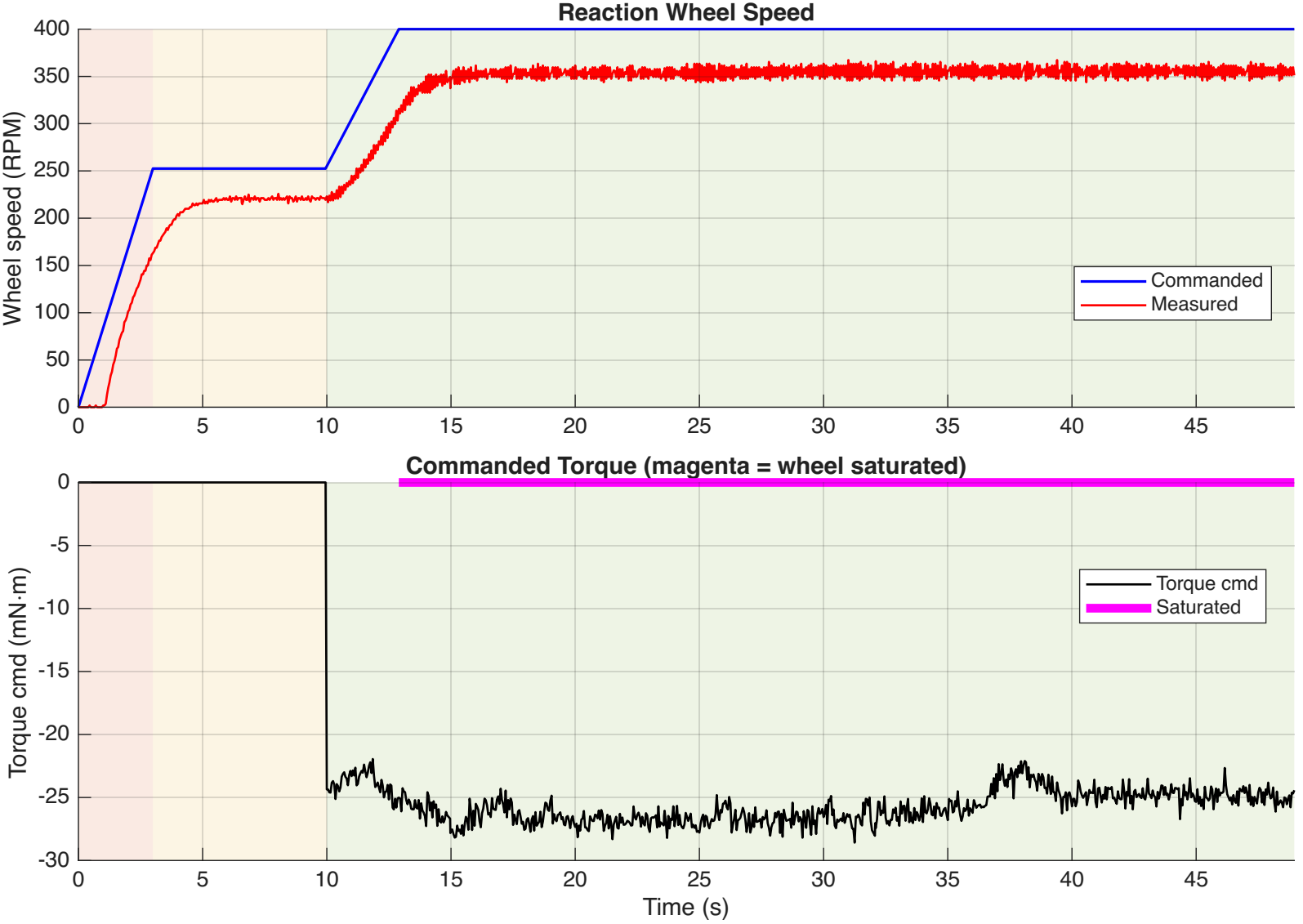
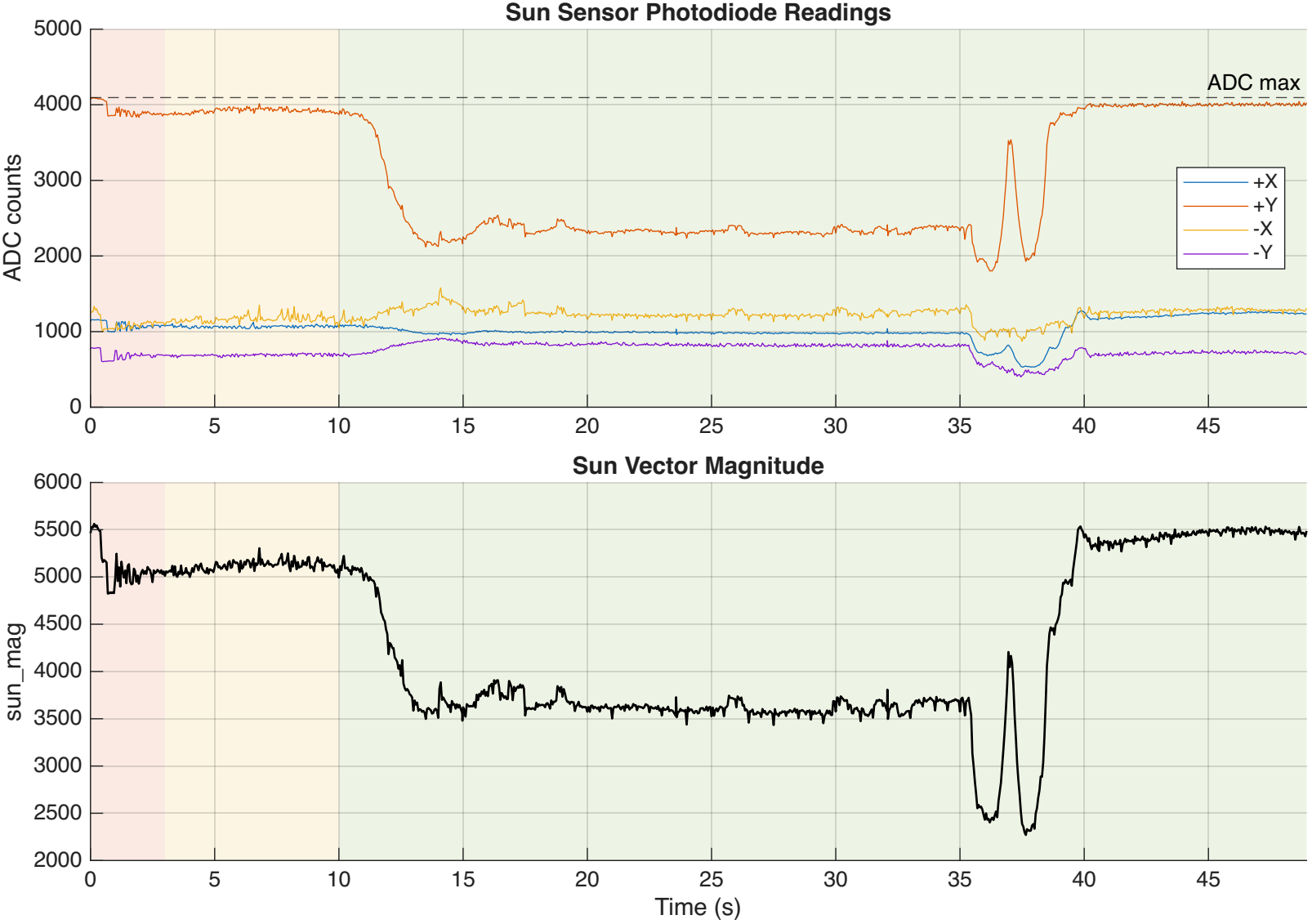
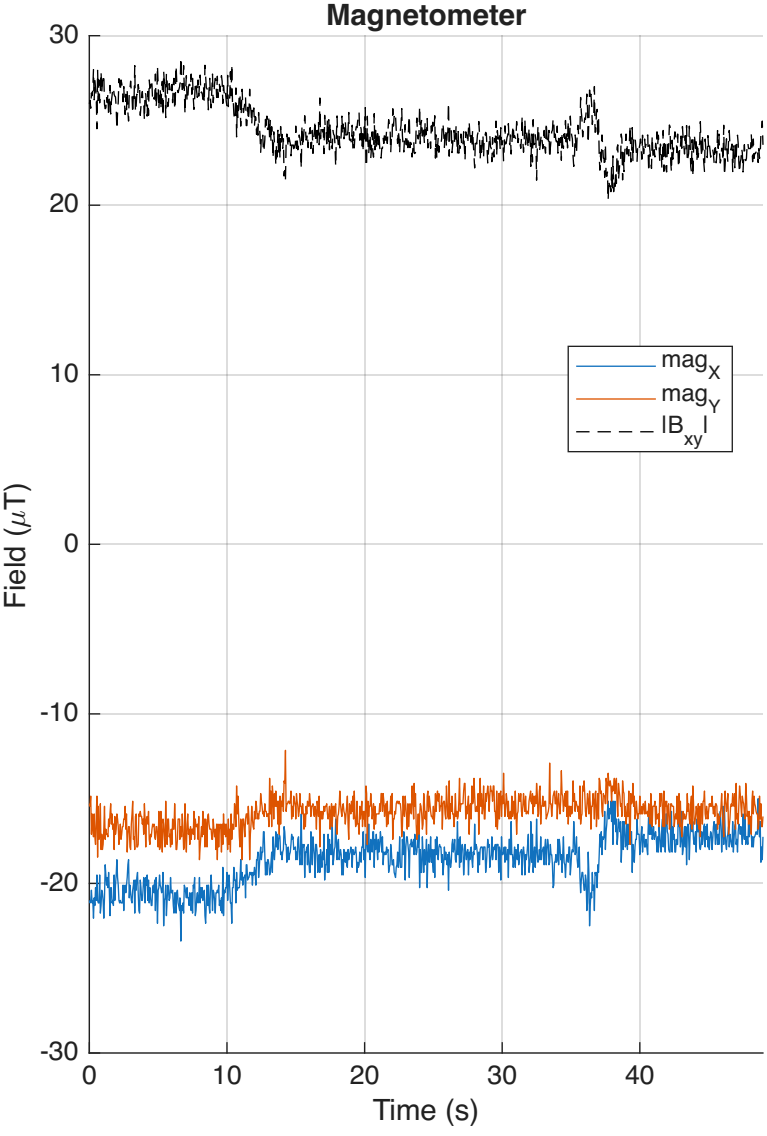
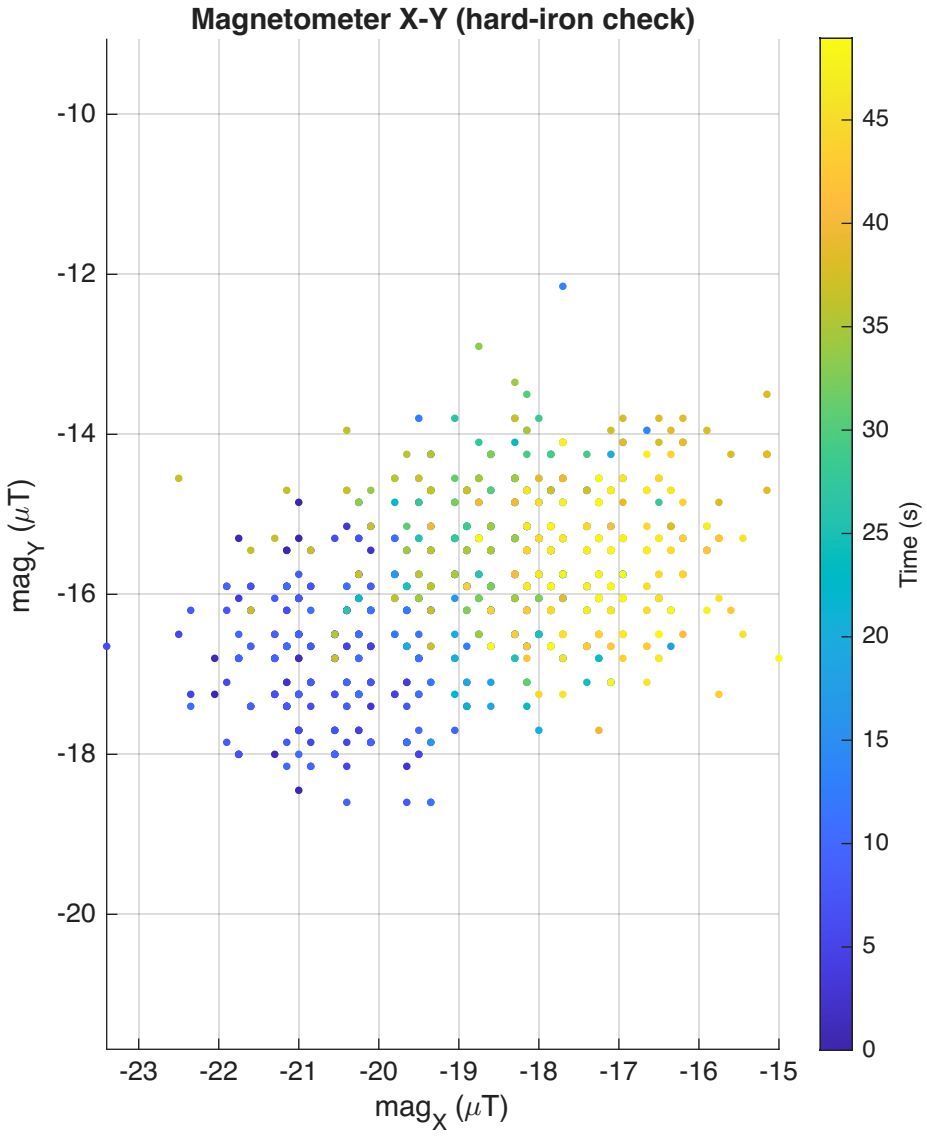

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
5
10
15
20
25
30
35
40
45
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

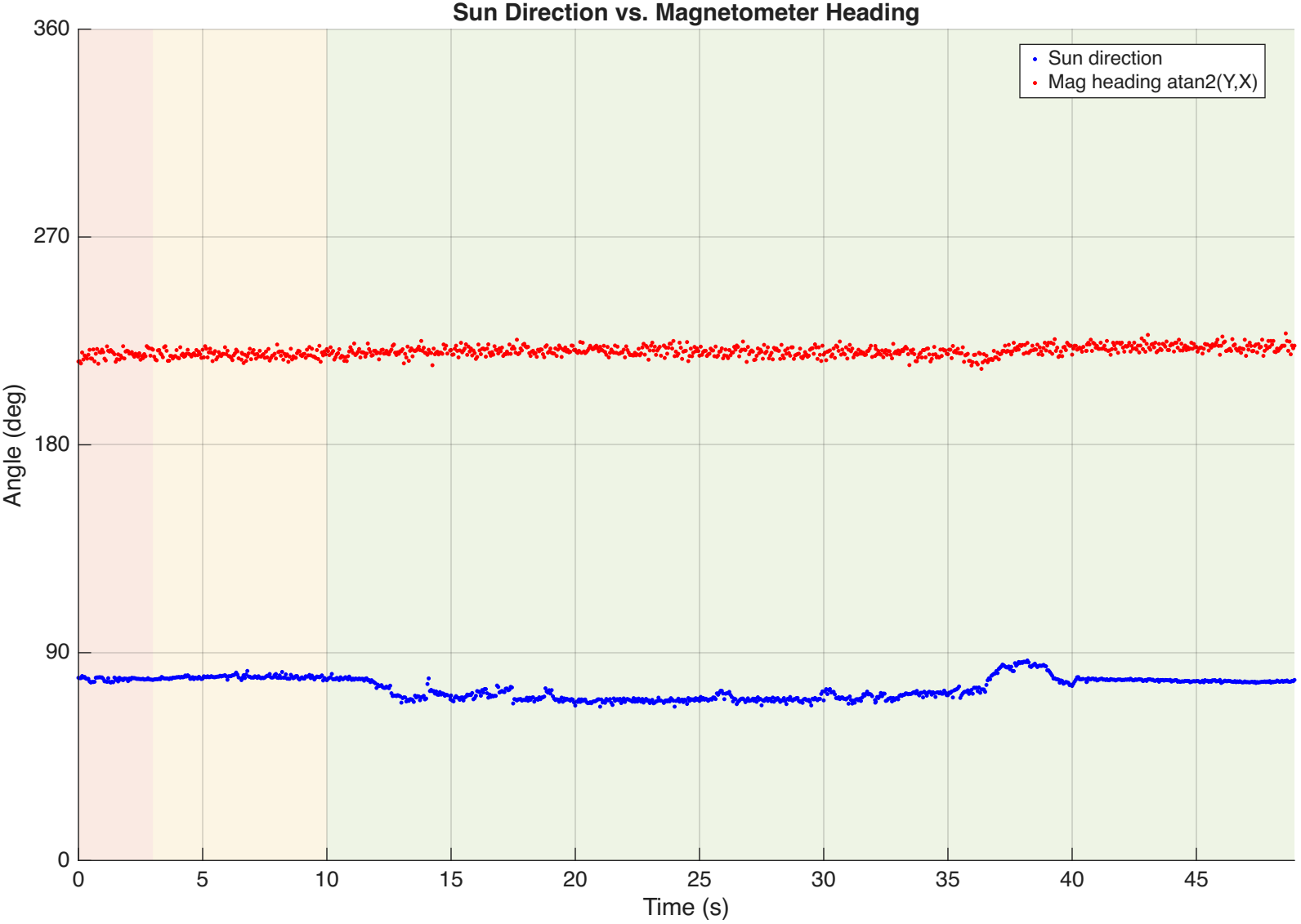

Z-Axis Body Rate (Gyro)
Body rate (deg/s)
Time (s)
5
0
-5
-10
0
5
10
15
20
25
30
35
40
45
SPINUP
HOLD
TRACK

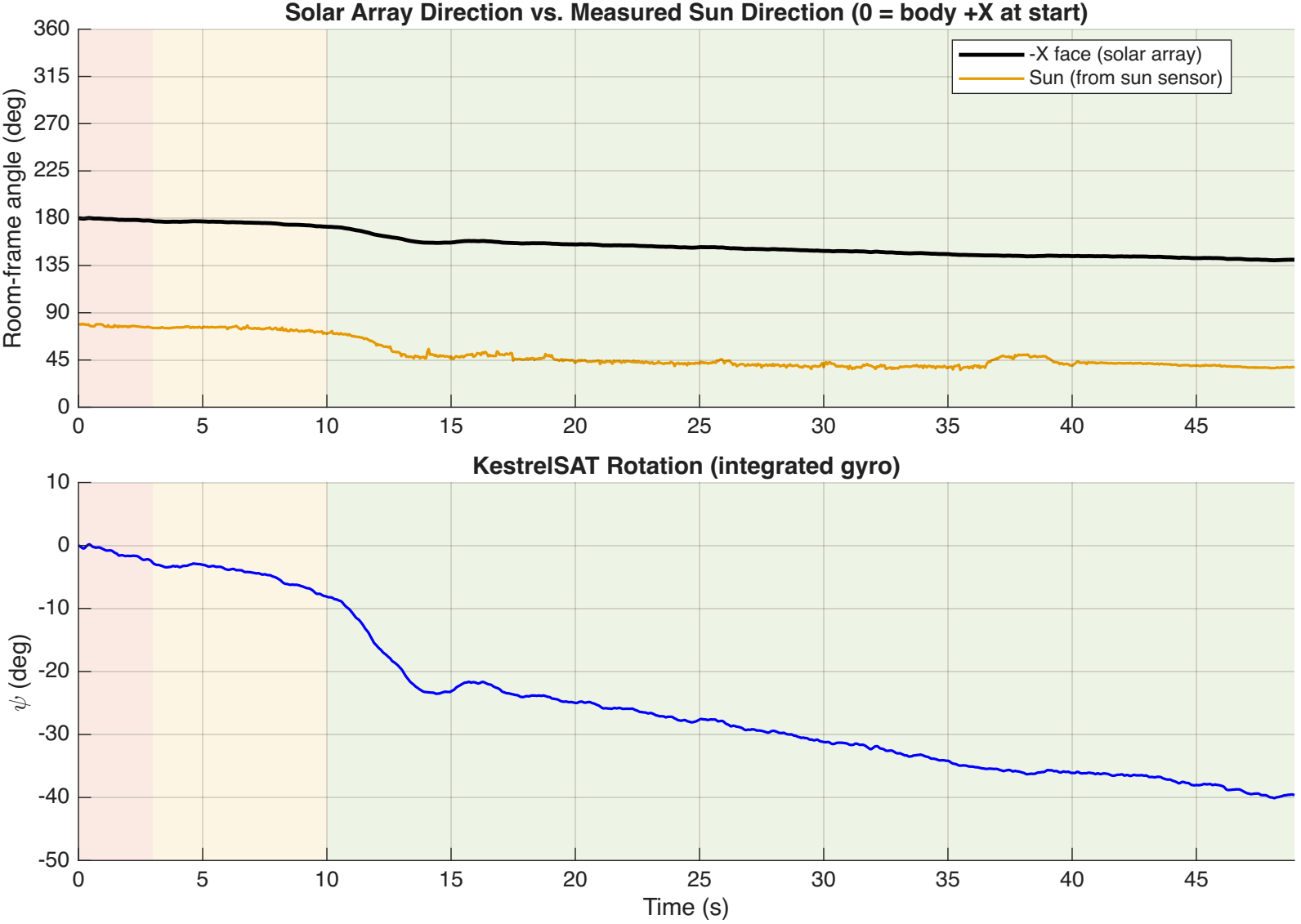

Reaction Wheel Speed
Wheel speed (RPM)
400
350
300
250
200
150
100
50
0
0
5
10
15
20
25
30
35
40
45
Commanded
Measured
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
0
-5
-10
-15
-20
-25
-30
Torque cmd
Saturated
Time (s)

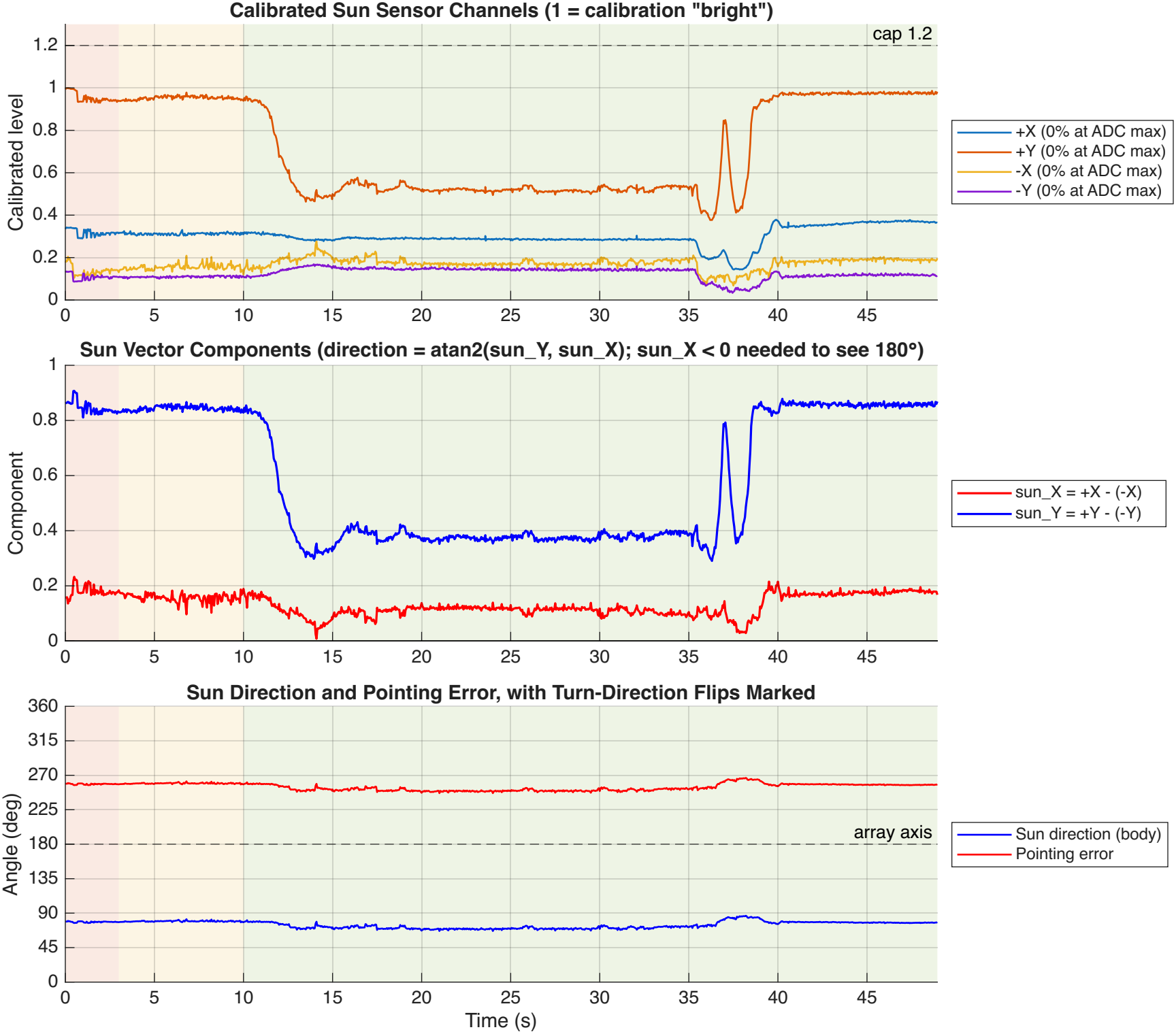

Sun Sensor Photodiode Readings
ADC max
+X
+Y
-X
-Y
ADC counts
5000
4000
3000
2000
1000
0
0
5
10
15
20
25
30
35
40
45
Sun Vector Magnitude
sun_mag
6000
5500
5000
4500
4000
3500
3000
2500
2000
Time (s)

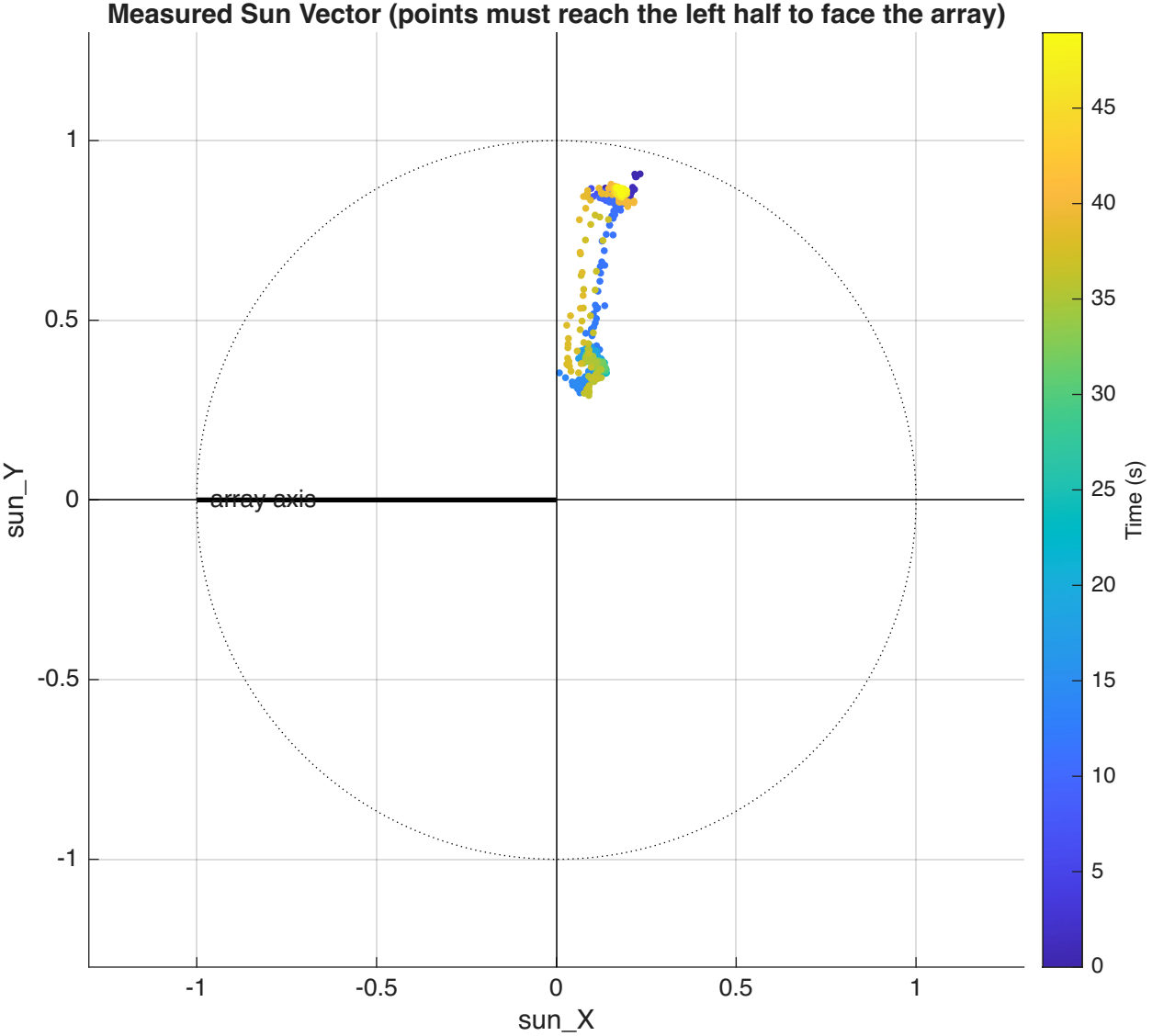

Magnetometer
30
20
10
0
-10
-20
-30
Field (μT)
0
10
20
30
40
Time (s)
mag$_X$
mag$_Y$
|B$_{xy}$|

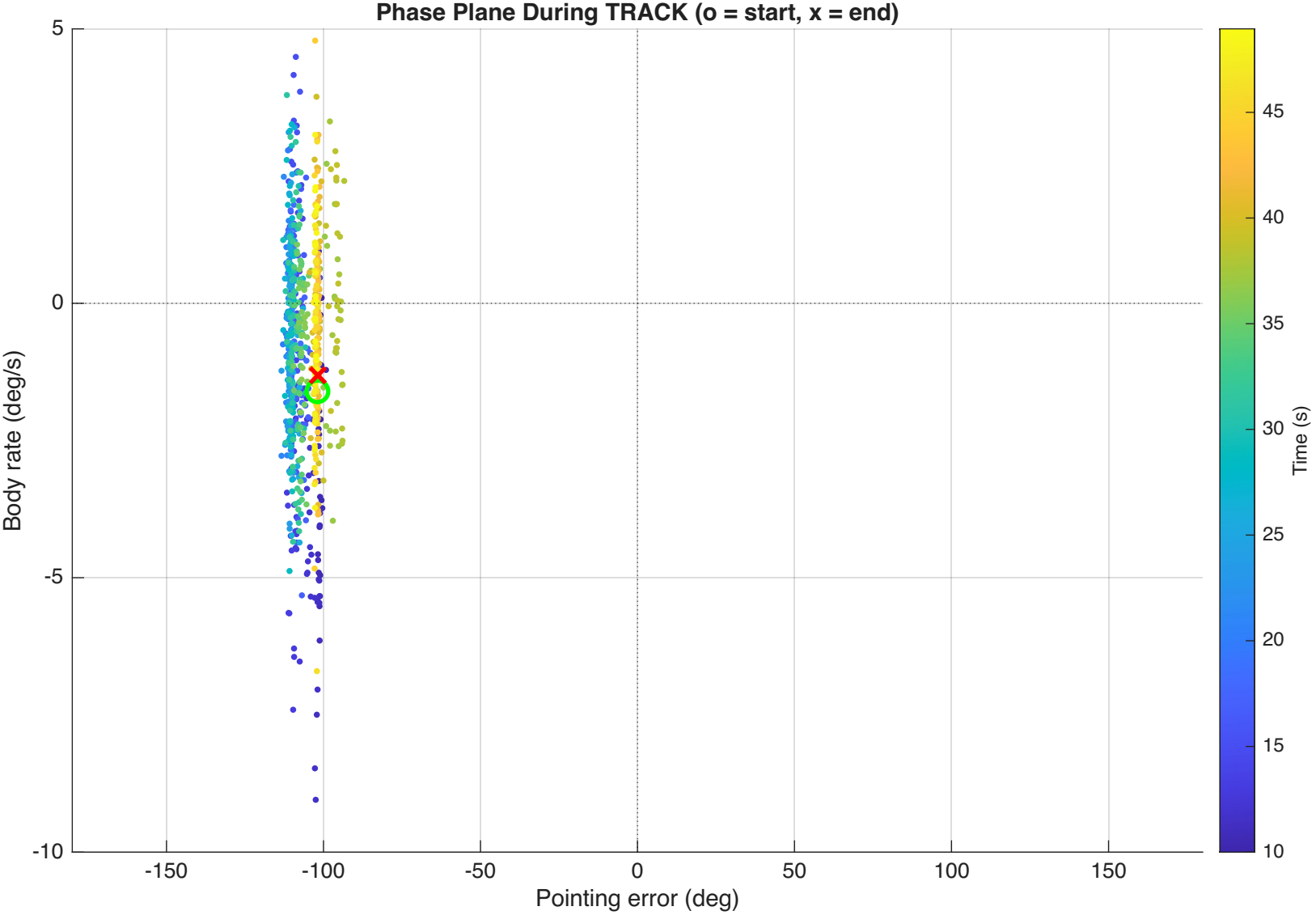

Magnetometer X-Y (hard-iron check)
-10
-12
-14
-16
-18
-20
mag$_Y$ (μT)
-23
-22
-21
-20
-19
-18
-17
-16
-15
mag$_X$ (μT)
45
40
35
30
25
20
15
10
5
0
Time (s)

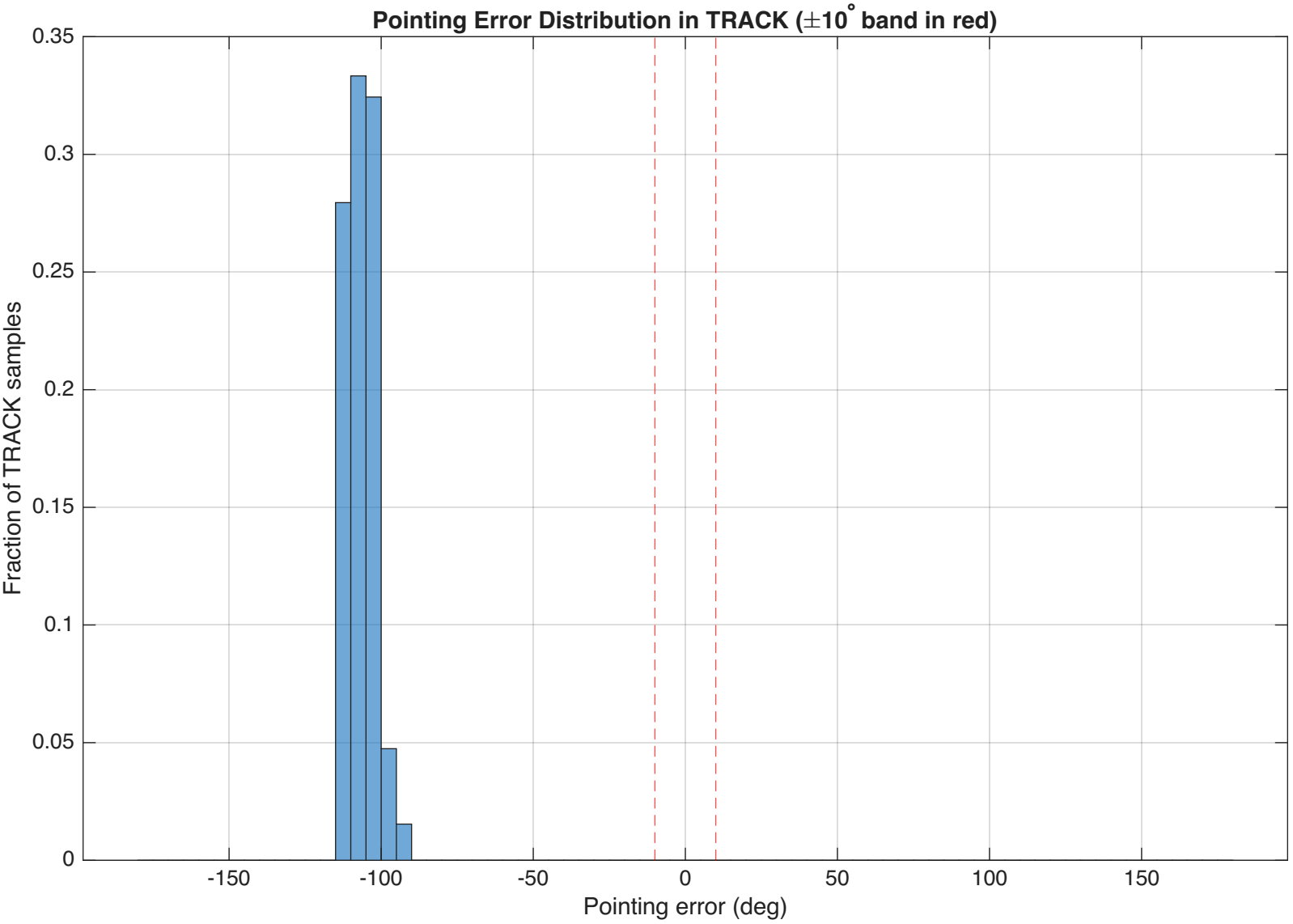

Sun Direction vs. Magnetometer Heading
Sun direction
Mag heading atan2(Y,X)
Angle (deg)
360
270
180
90
0
0
5
10
15
20
25
30
35
40
45
Time (s)


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
-X face (solar array)
Sun (from sun sensor)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
5
10
15
20
25
30
35
40
45
KestrelSAT Rotation (integrated gyro)
ψ (deg)
10
0
-10
-20
-30
-40
-50
Time (s)


Calibrated Sun Sensor Channels (1 = calibration "bright")
cap 1.2
Calibrated level
+X (0% at ADC max)
+Y (0% at ADC max)
-X (0% at ADC max)
-Y (0% at ADC max)
Sun Vector Components (direction = atan2(sun_Y, sun_X); sun_X < 0 needed to see 180°)
Component
sun_X = +X - (-X)
sun_Y = +Y - (-Y)
Sun Direction and Pointing Error, with Turn-Direction Flips Marked
Angle (deg)
array axis
Sun direction (body)
Pointing error
Time (s)


Measured Sun Vector (points must reach the left half to face the array)
array axis
sun_Y
sun_X
Time (s)
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
45
40
35
30
25
20
15
10
5
0


Phase Plane During TRACK (o = start, x = end)
Body rate (deg/s)
Pointing error (deg)
Time (s)
5
0
-5
-10
-150
-100
-50
0
50
100
150
10
15
20
25
30
35
40
45


Pointing Error Distribution in TRACK (±10° band in red)
Fraction of TRACK samples
Pointing error (deg)
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-150
-100
-50
0
50
100
150